Haojun Liang

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Research Interests:
Natural Language Processing, Large Language Models, Computational Linguistics

Education

- **Technical University of Munich** — Munich, Germany
  Bachelor of Science in Information Engineering — *Oct 2024 – Present*

Research Visits

- **The Chinese University of Hong Kong** – **Faculty of Engineering** — Hong Kong, China
  Visiting Student of HK PhD Fellowship Summer Workshop — *Jul 2026*
- **The Hong Kong Polytechnic University** – **Graduate School** — Hong Kong, China
  Visiting Student of International Research Summer School — *Jun 2026 – Jul 2026*
- **Renmin University of China** – **School of Information** — Beijing, China
  Research Assistant | Advisor: Prof. Wenxuan Wang — *Dec 2025 – Present*

Publications

* indicates equal contribution.

[P1] **Haojun Liang***, Ziwei Zhao*, Chenyu Li, Qiyue Zhao, Rongduomijia Mu, Yulan Shen, Yu Gu, Chen Zhang, Kairui Zhang, Yanbiao Ma, Wenxuan Wang. ScopeBench: Evaluating Cultural Norm Scope Awareness in LLMs. *In Proceedings of the 2026 Conference on Empirical Methods in Natural Language Processing (EMNLP 2026).*

[P2] Ziwei Zhao*, Yu Gu*, **Haojun Liang***, Chen Zhang, Xizhi Ding. Skill-as-API: Confidential Multi-Agent Coordination for Agentic Software Engineering. *KDD 2026 Workshop on Agentic Software Engineering (AgenticSE).*

Manuscripts in Preparation

[P3] **Haojun Liang**, Wenxuan Wang. CulturalPanel: Calibrating Cultural Scope Reasoning through Multi-Agent Disagreement. In preparation.

[P4] **Haojun Liang**, Bolei Ma. The Modality-Tense Interface in Large Language Models. In preparation.

Research Experience

- **Evaluating Large Language Models on Linguistic Competence [P4]** — LMU Munich
  Bachelor's Thesis | Advisor: Dr. Bolei Ma — *Aug 2026 – Present*
  - Investigating how LLMs distinguish temporal and modal interpretations of polysemous English auxiliaries such as *will*, *can*, *may*, *must*, and *should*, using corpus-based annotation and controlled minimal-pair evaluation.
  - Developing a two-level evaluation framework that combines behavioral tasks with layer-wise probing and control experiments to examine whether successful linguistic judgments are reflected in models' internal representations or driven by surface cues.
- **CulturalPanel: Disagreement-Aware Cultural Scope Reasoning [P3]** — Renmin University of China
  Project Lead | Advisor: Prof. Wenxuan Wang — *May 2026 – Present*
  - Developing CulturalPanel, a retrieval-grounded multi-agent framework that represents nine cultural clusters with agents grounded in cluster-specific corpora, aggregating their endorsement patterns to infer whether norms are universal, culture-specific, or contested.
  - Leveraging cross-agent disagreement as a calibration signal to identify unreliable scope judgments and trigger selective abstention or clarification; designed SCOPE-SWAP to evaluate over-universalization and over-relativization across cultural identities.

- **ScopeBench: Multilingual AI Cultural Safety Benchmark [P1]** Renmin University of China
  Project Lead | Advisor: Prof. Wenxuan Wang *Dec 2025 – Apr 2026*
  - Constructed ScopeBench, a benchmark of 3,323 cultural norms spanning 49 countries and nine cultural clusters, with externally anchored Universal/Specific scope labels validated through in-country annotators and cross-cultural human rights and ethics frameworks.
  - Designed a four-task evaluation framework and unified taxonomy of Scope Inflation, Scope Deflation, and Scope Displacement to trace cultural-scope failures from classification and cultural attribution to behavioral interaction; evaluated 21 frontier LLMs and found that near-ceiling scope classification does not translate into behavioral safety.
- **Multilingual Polarization Detection and Mitigation** Technical University of Munich | GitHub
  Bachelor Practical Course Project | Advisor: Dr. Daryna Dementieva *Oct 2025 – Feb 2026*
  - Constructed multilingual polarization benchmarks by prompting state-of-the-art LLMs as annotators, enabling systematic cross-lingual and cross-model evaluation of political polarization in generated text.
  - Designed and implemented a multi-agent mitigation pipeline integrating stance classification, argumentative feature extraction, and controlled rewriting for text neutralization and perspective diversification.

Open Source Projects

- **Skill-as-API: P2P AI Agent Skill Collaboration over XMTP [P2]** GitHub
  Project Contributor *Jan 2026 – Mar 2026*
  - Designed a Skill-as-API architecture where agent skills execute locally and results are delivered over E2E-encrypted XMTP channels using MLS, ensuring code, prompts, and domain knowledge remain private from callers across the open internet.
  - Implemented a four-tier trust system featuring automated trust decay and owner-controlled skill hiding, alongside an asynchronous delegation engine with a SQLite-persisted session state machine, schema-hash-based AgentCard caching enabling sub-2-second reconnects, and an MCP bridge supporting Claude Code and Cursor without runtime dependencies.
- **agent-synapse: Shared Decision Context for Parallel AI Agents** GitHub
  Project Developer *Mar 2026 – Apr 2026*
  - Built an MCP server in TypeScript that provides a shared SQLite-backed decision log for parallel AI agents, enabling agents to access prior architectural decisions involving database selection, API constraints, and dependency management before starting work, thereby reducing silent conflicts in multi-agent coding workflows.
  - Designed a structured decision schema with type classification, affected-path indexing, and supersession chains, while exposing three MCP tools including `write_decision`, `read_context`, and `query_decisions`, compatible with Claude Code, Cursor, and other MCP-compliant agent frameworks.
- **ctx-budget: Token Budget Manager for LLM Context Windows** GitHub
  Project Developer *Mar 2026 – Apr 2026*
  - Designed and implemented a priority-based token allocation engine that formulates LLM context window management as a resource scheduling problem, supporting pluggable strategies such as `PriorityGreedy` and `Proportional` through a clean abstract interface.
  - Built the system entirely with Python standard-library components, implementing token counting, overflow handling through truncation and dropping policies, and OpenAI-compatible message assembly from scratch, with 43 unit tests covering edge cases comprehensively.

Languages

Mandarin (native) | **Cantonese** (native) | **English** (professional working proficiency) | **German** (elementary)